

Replication Consistency and Collective Activity

What FIN coupling laws do not yet select

A conditional rigidity theorem, sharp finite counterexamples,
and an exact audit of the neural interpretation

Programmes ST8579–ST8590

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A twelve-part continuation of ST8549–ST8578. The main result constrains collective event rates through consistency under replication. It reduces one conditional independence test to same-origin pair activities, while showing why finite replication, maximum collective activity and a static neural weight matrix do not select a unique physical theory.

Abstract

For a projectively consistent, exchangeable family of finite-state Markov population generators with fixed singleton generator Q , let b_{ij} be the instantaneous rate at which both coordinates change when initially prepared at (i, j) . Arbitrarily large replication forces the matrix $B = (b_{ij})$ to be a completely positive real matrix, hence positive semidefinite. Consequently $b_{ii} = 0$ for every origin forces $B = 0$ and the independent tensor-sum generator at every population size. Small diagonal activities give linear, rather than merely square-root, population-error bounds under these premises.

A strict-FIN construction is consistent, exchangeable, reversible and dihedrally symmetric through population size three, with zero same-origin pair activity and positive mixed-origin activity. It cannot be extended consistently to size four. Thus the unlimited-replication premise is essential. Conversely, even maximal possible collective activity leaves different irreducible microscopic laws with the same singleton dynamics, equilibrium, symmetries and activity matrix.

The historical coupling diagrams and mirror operators do not supply the required population-composition law. An exact interval/integer audit also finds two fixed points and 111 two-cycles for a declared synchronous threshold network with strict weights. The canonical legacy network has six fixed points and 64 two-cycles. These correct an unsupported unique attractor interpretation; they are not physical vacua or solitons. No microscopic population, unit, selector, apparatus or ToE closure is derived. The next target is a sourced, transition-resolved composition law.

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1 Outcome, scope and the twelve investigations

The previous campaign proved that a specified singleton Markov generator plus absence of all simultaneous multi-coordinate events fixes the full population generator. Its principal limitation was the missing source of that absence. This continuation asks whether internal consistency can replace part of that premise. The answer is a useful conditional theorem, not an unconditional physical derivation.

Programme	Executed investigation and result
ST8579	Typed audit of kernel transformation, mirror coupling and neural updates: none of the inspected formulas specifies the required population event law. Historical statements are not imported as source theorems.
ST8580	Finite replication bound, including a target-resolved version.
ST8581	Unlimited replication plus zero same-origin activity forces the independent transition generator.
ST8582	Stronger activity-Gram theorem: complete positivity and quantitative constraints from repeated groups of prepared origins.
ST8583	An explicit finite-horizon counterexample and an obstruction to any four-copy extension, not merely to one implementation.
ST8584	Strict C_{12} replay on 144 and 1728 configurations, preserving singleton dynamics, reversal, equilibrium, graph symmetry and deletion.
ST8585	Diagonal-activity bounds propagated to full finite-population transition total variation.
ST8586	The signless graph Laplacian appears as a legitimate common-event activity Gram matrix. It is not the only realization of W .
ST8587	Even entrywise maximal activity does not select a unique microscopic law; two explicit irreducible alternatives remain.
ST8588	Full, sign-certified threshold-network catalogues for strict and legacy; a unique global-attractor interpretation is rejected.
ST8589	Excitation/inhibition and update-schedule audit, including a retained failed finite-time test and its symmetry-based explanation.
ST8590	Source-obligation synthesis and twelve ranked follow-up questions.

Proven means a displayed analytic argument or a finite exact certificate in its explicit model. Floating matrix checks are numerical validation, not interval proofs. *Refuted* concerns a stated implication, not the entire FIN programme. The interpretation of FIN as primordial fractal information is a motivating hypothesis, not laboratory evidence. No claim of global mathematical priority is made.

The reading focused on current AGENTS guardrails and the preceding campaign. Inspected sources included the kernel transformation diagram (`DIAGRAMS_KERNEL_TRANSFORMATION.md`), the post-41–50 correction, the negative-information-coupling report, relevant adaptive-law passages of the kernel referee report, and the complete `neural_coupling_investigation.py`. This is a targeted source audit, not another claim to have reverified every historical repository result. Historical files remain unchanged; corrections are recorded here and in AGENTS. Scientific tests and exact output certificates are packaged with the report.

2 Typed source audit: what the coupling formulas actually specify

The strict supplied weights, on $X = \{0, \dots, 11\}$, are

$$W_{ij} = \frac{\cos(0.18575d + 0.16250)}{1 + d^{9/5}}, \quad d = \min(|i - j|, 12 - |i - j|), \quad W_{ii} = 0,$$

with $A = sI - W \succeq 0$ and $s \simeq 1.6603072787660986$. The canonical legacy profile is instead

$$V_{ij} = 4 \log 2 \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d}.$$

It is signed and is not substituted as a classical rate matrix. We use it only in explicitly declared real symmetric spin-network tests below.

The historical four-mechanism diagram combines attenuation, resonance, oscillation and path summation into a scalar profile in d . It does not define an event $x \rightarrow y$ in X^N , a replication map, or a probability law for joint coordinate changes. Its arithmetic defect $d^{1.6}d^{-0.6} = d$, not a reciprocal damping law, was already corrected in the negative-information report and is not reopened as a new discovery.

The mirror construction $H_\lambda = H_0 + \lambda C$, with $RCR = -C$, types a Hermitian orientation-sensitive sector. It supplies neither a population jump schedule nor a theorem selecting the signed λ . A deterministic adaptive law $\dot{\psi} = iK\psi$, $\dot{K} = F(\psi, K)$ likewise requires F and its state space. It is not a specification of a stochastic generator on X^N . Hermiticity gives exact norm conservation in this state equation; it does not require adiabaticity and does not identify multi-particle noise.

Inspected object	Mathematical output	Missing input for the present theorem
Radial strict or legacy kernel	A finite weight profile	Population state space, allowed joint events and their rates
Mirror-odd C	A conditional Hermitian sector	Replication law and stochastic interpretation
Retention / negative feedback	A chosen rate modification or controller	Microscopic event composition, not merely damping strength
Hebbian / adaptive update	An operator–state differential or iterative law	Sampling, update schedule and the connection to singleton Markov paths
Fractal graph refinement	Finer vertices, collapse and possibly hidden rates	A theorem identifying vertex refinement with replication of constituents

This is a typing obstruction for the inspected formulas, not proof that no future FIN source law can be written. In particular, two operations must remain separate:

$$\underbrace{X \longrightarrow X_{\text{fine}}}_{\text{resolution of one carrier}} \neq \underbrace{X \longrightarrow X^N}_{\text{population composition}}.$$

The latter does not introduce an informational substrate below the nadsoliton. It is a candidate description of a composite system, whose physical existence and interpretation have not been derived. “Clonal” preparation below means classical coordinates prepared at the same known vertex, not copying an arbitrary unknown quantum state.

3 Finite replication: a rate-counting theorem

Let G_N be finite-state continuous-time Markov generators on X^N . Assume exchangeability under coordinate permutations and strong projective consistency under deleting any coordinates. In particular every singleton has the same generator Q at every configuration, not only at equilibrium. Write $r_i = -Q_{ii} < \infty$. For two coordinates initially at (i, j) define

$$b_{ij} = \sum_{a \neq i, b \neq j} G_2((i, j), (a, b)), \quad \delta_i = b_{ii}, \quad (1)$$

$$\beta_{ij}^b = \sum_{a \neq i} G_2((i, j), (a, b)), \quad b \neq j. \quad (2)$$

Exchangeability gives $b_{ij} = b_{ji}$. The diagonal b_{ii} concerns two different coordinates that share an initial vertex; it is *not* a single-coordinate self-loop Q_{ii} or W_{ii} .

Theorem 1 (Finite same-origin bound). *If the family extends consistently through $m+1$ coordinates and $\delta_i = 0$, then*

$$b_{ij} \leq \frac{r_j}{m}, \quad \beta_{ij}^b \leq \frac{Q_{jb}}{m}.$$

Proof. Prepare m coordinates at i and one at j . Strong consistency makes the joint-change rate of any pair among the i coordinates equal to $\delta_i = 0$. Positivity therefore excludes every event changing two or more of these m coordinates. Let Z count changed i coordinates in an event, and Y indicate whether the j coordinate changes. Thus $ZY \leq Y$. Summing over positive event rates gives $mb_{ij} \leq r_j$. Replacing Y by the indicator of the specific transition $j \rightarrow b$ proves the target-resolved bound. $\square$

Proposition 2 (Nonzero same-origin activity). *For $r_j > 0$ and $\delta_i \geq 0$, the same construction gives*

$$b_{ij} \leq F_m(r_j, \delta_i) := \frac{r_j}{2m} \left(1 + \sqrt{1 + \frac{4m(m-1)\delta_i}{r_j}} \right).$$

If $r_j = 0$, then $b_{ij} = 0$ directly.

Proof. Let $\langle \cdot \rangle_g$ mean summation over the event rates in that prepared configuration. Consistency gives

$$\langle ZY \rangle_g = mb_{ij}, \quad \langle Y \rangle_g = r_j, \quad \langle Z(Z-1) \rangle_g = m(m-1)\delta_i.$$

Cauchy–Schwarz in the positive measure restricted to $Y = 1$ gives $(mb_{ij})^2 \leq r_j \langle Z^2 Y \rangle_g$. Also $\langle Z^2 Y \rangle_g \leq m(m-1)\delta_i + mb_{ij}$. Solving the resulting quadratic inequality proves the bound. $\square$

For zero δ_i this reduces to r_j/m . For positive δ_i its limit is $\sqrt{r_j \delta_i}$. The next theorem improves that one-group estimate by replicating *both* origins, or all origins together.

4 The activity-Gram theorem

Theorem 3 (Replication activity matrix). *Let $B = (b_{ij})$ and let $n = |X|$. If a consistent family exists through nm coordinates, then*

$$B + \frac{1}{m} \text{diag}(r_i - \delta_i) \quad \text{is a completely positive real matrix.} \quad (3)$$

If it extends to arbitrarily large population sizes, B itself is a completely positive real matrix. In particular,

$$B \succeq 0, \quad 0 \leq b_{ij} \leq \sqrt{\delta_i \delta_j}.$$

Here “completely positive matrix” means ZZ^T for an entrywise nonnegative real matrix Z ; it is not terminology for a quantum channel.

Proof. Prepare m coordinates at each origin i . For one event let Z_i count the changed coordinates in group i . These are nonnegative integers. The matrix

$$M_m = \frac{1}{m^2} \sum_{y \neq x} g(x, y) Z(x, y) Z(x, y)^T$$

is a sum of nonnegative rank-one outer products. Projective consistency gives

$$\langle Z_i \rangle_g = m r_i, \quad \langle Z_i(Z_i - 1) \rangle_g = m(m - 1) \delta_i, \quad \langle Z_i Z_j \rangle_g = m^2 b_{ij} \quad (i \neq j).$$

These identities yield exactly (3). Taking arbitrarily large m gives the limit B .

For completeness, the real completely-positive cone is closed here. In the finite-dimensional space of symmetric matrices, any conic combination can be reduced by linear dependence to at most $n(n + 1)/2$ nonnegative rank-one terms. Normalize each nonzero generating vector to unit norm. The coefficients then sum to the matrix trace. Along a convergent sequence the coefficients are bounded, the vectors lie in a compact nonnegative part of the unit sphere, and a subsequence of these finite representations converges to one of the same form. Thus B has such a representation. Its 2×2 principal minors give $b_{ij}^2 \leq \delta_i \delta_j$. $\square$

For just two origins, $2m$ copies suffice for the corresponding principal matrix. Hence a finite family through M copies gives

$$b_{ij} \leq \sqrt{\left(\delta_i + \frac{r_i - \delta_i}{m}\right) \left(\delta_j + \frac{r_j - \delta_j}{m}\right)}, \quad m = \lfloor M/2 \rfloor.$$

This can be combined with the one-group bound rather than replacing it: the latter can be sharper at small horizons or for a specified target. For equal $r_i = s$ and $\delta_i \leq \delta \leq s$, the two-group bound is $\delta + (s - \delta)/m$, and the infinite-horizon bound is simply δ .

Corollary 4 (Conditional independence from same-origin tests). *For an exchangeable strongly projectively consistent family at arbitrarily large sizes, $b_{ii} = 0$ for every i forces $G_N = \sum_{k=1}^N Q^{(k)}$ for every finite N .*

Proof. The Gram bound forces every b_{ij} to vanish. In any N -coordinate configuration, strong consistency makes the pair-change rate for each coordinate pair equal to its corresponding b_{ij} . Every event changing two or more coordinates contributes positively to some pair rate and is therefore absent. Only single-coordinate jumps remain; autonomous singleton rates fix each of their rates to Q . Row sums fix the diagonal. $\square$

This reduces the *conditional* activity test to $|X|$ same-origin preparations, but it imports an unlimited-replication premise. No finite experiment verifies that premise for every size. A FIN derivation must establish the composition rule mathematically. The theorem does not say that independence is physically preferable, and a valid FIN law might instead predict nonzero joint activity.

4.1 Quantitative propagation to the previous theorem

For any finite configuration let

$$\mathcal{B}_N(x) = \sum_{y \neq x} g(x, y) \binom{\#\{k : y_k \neq x_k\}}{2}.$$

Consistency gives $\mathcal{B}_N(x) = \sum_{k < l} b_{x_k x_l}$. Under unlimited replication and $\max_i \delta_i \leq \delta$,

$$\sup_x \mathcal{B}_N(x) \leq \binom{N}{2} \delta.$$

The preceding campaign’s finite-generator perturbation argument therefore gives

$$\sup_x \text{TV} \left(e^{tG_N}(x, \cdot), e^{t \sum_k Q^{(k)}}(x, \cdot) \right) \leq \min\{1, N(N-1)\delta t\}.$$

For reference, the perturbation proof replaces each joint event changing k coordinates by its k marginal single events. The row total-variation norm is at most $\sum kg \leq 2\mathcal{B}_N$. Duhamel’s formula and stochastic contraction then give $2t \sup \mathcal{B}_N$. This bounds transition laws, not complete path records. Initial correlations, detector calibration and the physical clock remain separate.

5 A sharp finite-horizon strict-FIN counterexample

Let $X = C_{12}$, $f(i) = i + 6 \pmod{12}$, and let the two parity classes be even and odd vertices. The antipodal map preserves parity. Put $a = W_{i,f(i)} = W_6$ and fix a proposed maximum population size M and

$$c = \frac{a}{M-1}.$$

For each $N \leq M$ use the following rates.

1. Coordinate k makes each non-antipodal transition with its original rate $W_{x_k j}$.
2. Its antipodal rate is $a - cn_k^{\text{opp}}$, where n_k^{opp} is the number of other coordinates in the opposite parity class.
3. Each unordered pair of coordinates in opposite parity classes makes the simultaneous antipodal move at rate c .

Theorem 5 (Finite consistency without infinite extendibility). *This defines a nonnegative, exchangeable, strongly projectively consistent family through size M . Its singleton generator is exactly strict Q . It is reversible with the same product-uniform equilibrium and has the full D_{12} symmetry of the supplied graph. It is irreducible. Its pair activities satisfy*

$$b_{ij} = \begin{cases} c, & i, j \text{ have opposite parity,} \\ 0, & \text{otherwise,} \end{cases} \quad b_{ii} = 0.$$

For $c > 0$ fixed as above, no family satisfying the same premises and the same two-coordinate generator can extend through size $M + 1$.

Proof. Nonnegativity follows from $n_k^{\text{opp}} \leq N - 1 \leq M - 1$. The total antipodal rate of a coordinate is $a - cn_k^{\text{opp}} + cn_k^{\text{opp}} = a$. Deleting a partner converts its joint event, if active, into an observed singleton antipodal event; this exactly compensates the deleted partner’s subtraction from the singleton rate. All other rates restrict unchanged. This proves strong consistency, not just agreement of a numerical marginal.

An antipodal move preserves all parity labels. Its reverse therefore has the same partner counts and rate. Each common antipodal map is an involution. Non-antipodal rates are symmetric, so the full generator is symmetric and product-uniform measure is reversible. Non-antipodal single-coordinate edges include the connected nearest-neighbour cycle and remain positive; hence all configurations communicate. Particle permutations preserve the construction. Vertex rotations and reflections commute with the antipode and preserve the relation “opposite parity”, even when they exchange the two parity classes; hence there is no selected parity label or orientation.

For origins $(0, 1)$ the target-resolved joint rate for the second coordinate to make $1 \rightarrow 7$ is $c = a/(M - 1)$. If an extension to $M + 1$ existed, the finite same-origin theorem with $m = M$ would instead require $c \leq a/M$. This is a strict contradiction. It excludes every extension with the stated premises, not merely the displayed formula beyond its domain. $\square$

For $M = 3$,

$$a \simeq 0.01107081732144, \quad c \simeq 0.00553540866072.$$

The implemented G_2, G_3 have 144 and 1728 states. The deletion residual is about 2.66×10^{-15} , singleton residual 8.88×10^{-16} , and reversibility residual zero. All 24 dihedral transformations were checked. The forbidden fourth-copy target bound is $a/3 < c$. The algebraic proof, not the decimal residuals, licenses these conclusions.

The activity matrix has zero diagonal and positive opposite-parity blocks. For $v_i = (-1)^i$,

$$v^T B v = -72c < 0.$$

It therefore also violates the necessary infinite-replication Gram condition. A finite consistency check, even together with equilibrium and symmetry, must not be silently promoted to unlimited self-similarity.

6 A new useful object, but no unique “maximum resonance” law

The common-vertex-transposition model from the previous campaign has activity matrix

$$B_\theta = \theta(sI + W), \quad 0 \leq \theta \leq 1.$$

Indeed, two coordinates at the same origin i leave together whenever a common transposition touches i , giving diagonal rate θs . Two different origins change together only when their connecting pair is transposed, giving θW_{ij} .

This is the signless Laplacian, with an explicit nonnegative incidence factorization:

$$sI + W = \sum_{i < j} W_{ij} (e_i + e_j)(e_i + e_j)^T.$$

By comparison,

$$A = sI - W = \sum_{i < j} W_{ij} (e_i - e_j)(e_i - e_j)^T.$$

The same edge data therefore admit an oriented difference form for mean diffusion and an unsigned Gram form for one collective-event realization. This is standard graph algebra with an explicit FIN application, not a new physical duality theorem. The event construction is what gives the plus form its activity interpretation, and θ is still free. In particular, $W_{ii} = 0$ does *not* imply $b_{ii} = 0$.

One might try to select the missing dynamics by maximizing collective activity, as a precise proxy for the informal “maximum resonance” idea. Even that proxy does not determine the microscopic process.

Theorem 6 (Maximal-activity ambiguity). *There are distinct irreducible, reversible, exchangeable, strongly projectively consistent families with singleton generator Q , all strict graph symmetries, the same product-uniform equilibrium, and the same entrywise maximal activity matrix $B = s\mathbf{1}\mathbf{1}^T$.*

Proof. Set $R = W/s$, which is symmetric, stochastic and zero on the diagonal. Define

$$G_N^{\text{prod}} = s(R^{\otimes N} - I), \quad (G_N^{\text{shift}} h)(x) = \sum_{r=1}^{11} W_{0r} [h(x + r\mathbf{1}) - h(x)].$$

The first model has one rate- s Poisson clock, followed by conditionally independent vertex transitions at that tick. The second has common cyclic translations. Every coordinate changes at every event in either model, so every $b_{ij} = s$. This is the maximum possible rate since a pair-change event is contained in either coordinate’s rate- s departure event.

Both generators restrict to Q on one coordinate and restrict consistently under deletion. They are exchangeable. Symmetry of R and the equality of opposite-shift rates give reversibility,

while circulant structure gives the stated graph symmetries. For each $0 \leq \alpha < 1$ the mixture $(1 - \alpha)G_N^{\text{prod}} + \alpha G_N^{\text{shift}}$ is irreducible: R^2 is entrywise positive on this complete graph, so two product ticks can reach every configuration. All these mixtures retain the same B .

From $(0, 0)$, however, the product model has transition rate $W_{01}W_{02}/s > 0$ to $(1, 2)$, while a common shift has zero rate for that transition. Thus different mixtures have different generators. $\square$

For mixture weights $\alpha = 0.25, 0.75$, the strict target rates are about 0.04077154 and 0.01359051, although the entire activity matrix is identical. Hence the observed number of simultaneous changes can still omit *which* changes occur. A future source law must address the transition-resolved object

$$T_{(i,a),(j,b)} = G_2((i,j), (a,b)), \quad a \neq i, \quad b \neq j,$$

and, potentially, higher-order composition. Its further classification is recommended work, not a result claimed in this campaign.

7 Retained falsification: an invisible finite-time exception

An attempted regression test expected every mixed-event model in the finite construction to differ from independence in endpoint distributions after pure same-origin preparation. This expectation failed for a uniform four-state kernel. The failure was investigated, not erased by weakening the numerical tolerance.

Take $X = C_4$, $W_{ij} = 1$ off the diagonal, antipode $i + 2$, and the finite construction above. The singleton solution from origin i is

$$p_t = u + e^{-4t}(e_i - u).$$

The parity class not containing i has equal probabilities at its two vertices. In the opposite-parity block, the correction to the independent forward generator applied to $p \otimes p$ is proportional to

$$[p(f(a)) - p(a)][p(f(b)) - p(b)].$$

One factor vanishes. Outside that block the correction is zero. Thus $p_t \otimes p_t$ solves both forward equations from every pure clone preparation at every time. The uniqueness of the finite linear initial-value problem proves exact endpoint equality. Mixed-origin preparations instead distinguish the processes. This is not equality of all conditional paths or event records.

The strict weights break this special within-class balance. For strict preparation $(0, 0)$ at time 0.4, the numerically computed endpoint TV difference is about 0.0006491751, despite zero joint-change rate at the initial clone configuration. Both the uniform exception and the strict distinction remain in the test suite.

8 Exact neural audit: schedule, fixed points and false criticality

The inspected older script labels a few signs of eigenvectors as attractors, uses the noncyclic distance $|i - j|$, and infers a single global basin and physical vacuum stability. These inferences are unsupported by that test. The matrix alone does not specify a neural activation, update schedule, noise law, learning rule or physical interpretation.

8.1 A complete declared finite test

We explicitly supply the synchronous zero-threshold rule

$$x'_i = \text{sign}((Wx)_i), \quad x \in \{-1, 1\}^{12},$$

keeping the old spin if the field is zero. We enumerate all 4096 configurations for strict cyclic, canonical legacy cyclic and legacy noncyclic weights. No tested field is zero, as certified below.

Declared network	Fixed points	Two-cycles	Longer cycles
Strict, cyclic distance	2	111	0
Legacy, cyclic distance	6	64	0
Legacy, line distance	6	18	0

These are exact finite catalogues for the specified transcendental weights, not counts obtained merely by treating floats as exact. They do not count learned memories, physical vacua, particles or solitons. The additional floating energy data in the JSON are separately labelled numerical.

8.2 How the signs are certified

The checker encloses strict cosine values by alternating rational Taylor sums at the exact decimal-rational arguments. It encloses $d^{9/5}$ by an integer fifth-root search on a scaled version of d^9 . Legacy cosines are written exactly using $\sqrt{2}, \sqrt{3}, \sqrt{6}$ and their eight-step period; these square roots are enclosed by integer root searches. The amplitude uses the convergent positive series

$$\log 2 = 2 \sum_{k=0}^{\infty} \frac{(1/3)^{2k+1}}{2k+1}.$$

After 25 terms its remaining tail is bounded above by $2(1/3)^{51}/[51(1-1/9)]$.

Outward rounding gives integer matrices L, U with $L/10^{12} \leq W \leq U/10^{12}$ entrywise. For each spin configuration, the field lies in the exact interval with midpoint numerator $x^T(L+U)$, radius numerator given by the appropriate row sums of $U-L$, and denominator $2 \cdot 10^{12}$. All operations are bounded int64 operations; the maximum possible sums are checked far below overflow. All 49,152 field signs per matrix are separated from zero. Certified minimum field magnitudes are respectively greater than

$$\frac{2767704329}{250000000000}, \quad \frac{5294143023}{500000000000}, \quad \frac{464196229}{500000000000}.$$

The exact successor graph therefore belongs to the actual specified weights. Its complete cycle enumeration is checked independently by a strongly connected-component calculation for strict. The certificate contains the outward integer weight matrices and sign margins. This is not a proof-assistant verification; it is an explicit finite interval and integer certificate with documented analytic enclosures.

8.3 A general obstruction to one signed-spin fixed point

For any real symmetric zero-diagonal W , the finite energy $E(x) = -x^T W x / 2$ has a minimizer. Flipping spin i changes the energy by $2x_i(Wx)_i$. At a global minimum this is nonnegative, so that state is a fixed point of the threshold rule with the stated tie convention. Since $E(-x) = E(x)$, its distinct negative is also a fixed point. A unique fixed state is therefore impossible in this unquotiented model. For strict positive connected weights, the only ground configurations are the two uniform spin signs. Introducing a sign quotient or bias would be a new model choice, not something established by the old script.

8.4 Update schedules and excitation/inhibition are separate data

A single asynchronous alignment update never increases this energy: if a spin flips against its local field, $2x_i(Wx)_i \leq 0$. A simultaneous update need not decrease it, and the certified two-cycles already refute global convergence to one fixed point. Thus a single-coordinate update schedule builds absence of simultaneous changes into the algorithm; it does not derive that absence from the weight matrix. The original Hopfield paper likewise specifies an evolution algorithm rather than identifying arbitrary static matrices with complete dynamics [2].

Finally, an excitation/inhibition weight-sum ratio cannot determine linear stability. For the supplied continuous extension

$$\dot{z} = -z + gVz,$$

the largest growth eigenvalue is $-1 + g\lambda_{\max}(V)$. For any nonzero symmetric zero-diagonal V , trace zero implies $\lambda_{\max}(V) > 0$. Choosing $g = 1/(2\lambda_{\max})$ or $2/\lambda_{\max}$ gives maximum growth $-1/2$ or $+1$, with exactly the same matrix and E/I ratio. The canonical legacy line ratio is approximately 0.59904 in both cases. A leak, activation slope and gain must be specified before “critical” or “stable” has a defined dynamical meaning. This does not transfer a linear-flow result to the binary threshold dynamics or to physical matter.

9 Dual dynamics and the remaining source obligations

Nothing here changes the strict Dirichlet identity or the dual calculus

$$A = sI - W, \quad U_t = e^{-itA}, \quad P_t = e^{-tA}.$$

They remain different functions of one self-adjoint operator, with different operational semantics. The new activity matrices belong to a declared classical population extension; they do not equate heat with Born probabilities, derive a bath, or select a coherent quantum completion.

The constructive advance is conditional: unlimited exchangeable replication forces an activity Gram structure, and zero diagonal then forces independence. Its premises are not yet strict-sourced. Finite replication does not suffice, and maximizing activity does not uniquely select the alternative. We therefore neither impose independence nor reject self-coupling: the actual theory must determine its activity and transition laws.

Known exchangeable-process results permit common stochastic influences [1], so consistency should not be equated with independence in general. The proofs here are elementary finite-generator arguments and do not rely on an unverified application of a general representation theorem. No claim is made that these general mathematical mechanisms are newly discovered in the literature. Their role is to sharpen FIN’s source tests.

10 Twelve recommended next studies

The ordering is by concrete information value and local tractability, not by an invented probability of proving a Theory of Everything.

No.	Next test and acceptance criterion
1	Derive a transition-resolved composition law from an actual FIN coupling rule. It must distinguish the two maximal-activity families above, not merely reproduce W or B .
2	Define the directed-transition coincidence matrix T and prove which Gram, marginal and positivity conditions its replication consistency requires. Do not assume the aggregate B is informationally complete.
3	Test whether equality of full two-coordinate dynamics fixes higher coordinate laws. Seek exact third-order counterexamples before claiming pairwise closure.
4	Classify finite extendibility on small carriers using exact rational linear constraints and nonnegative rates; compare with the sharp target-resolved bound, not only matrix eigenvalues.

No.	Next test and acceptance criterion
5	Produce a strict-sourced map, if any, relating fractal refinement to population replication. Reject a map justified only by a common number of vertices or a visual self-similarity.
6	Optimize finite-horizon diagonal-to-pair bounds using both one-group and multi-group moment constraints. Export a verified certificate for any claimed optimal bound.
7	Extend the previous finite-resolution counting test to same-origin preparations with calibrated efficiency. Separate the testable finite data from the non-empirical unlimited-replication premise.
8	Design the minimum mixed-origin or transition-resolved test distinguishing the maximal-activity family. Include the uniform four-state endpoint alias as an adversarial control.
9	Audit the actual adaptive neural rule, with its activation, leak, sampling and schedule fixed. Recompute basins rather than testing only eigenvector signs.
10	Study whether a sourced memory or feedback mechanism changes the Markov composition class. A non-Markov escape is allowed, but requires a new theorem rather than importing the present Markov conclusion.
11	Compare the oriented and unsigned incidence constructions under an explicit legacy-to-strict map. Do not treat the new signless activity matrix as a replacement kernel or physical-role transfer.
12	Formalize the finite counting and Gram identities and replay the exact threshold certificates with an independent integer implementation; do not describe that as independent experimental evidence.

11 Verification and final judgment

The package contains the scientific computation, rational enclosure checker, 25 regression tests, exact sign certificates, numerical results, source audit ledger, figures and this English report. The tests include the retained uniform-kernel exception, exact small-matrix projection, strict sparse projection on 1728 states, symmetry checks, rational inequalities and high-precision cross-checks of transcendental enclosures. Tests do not supply physical units or an external scientific audit.

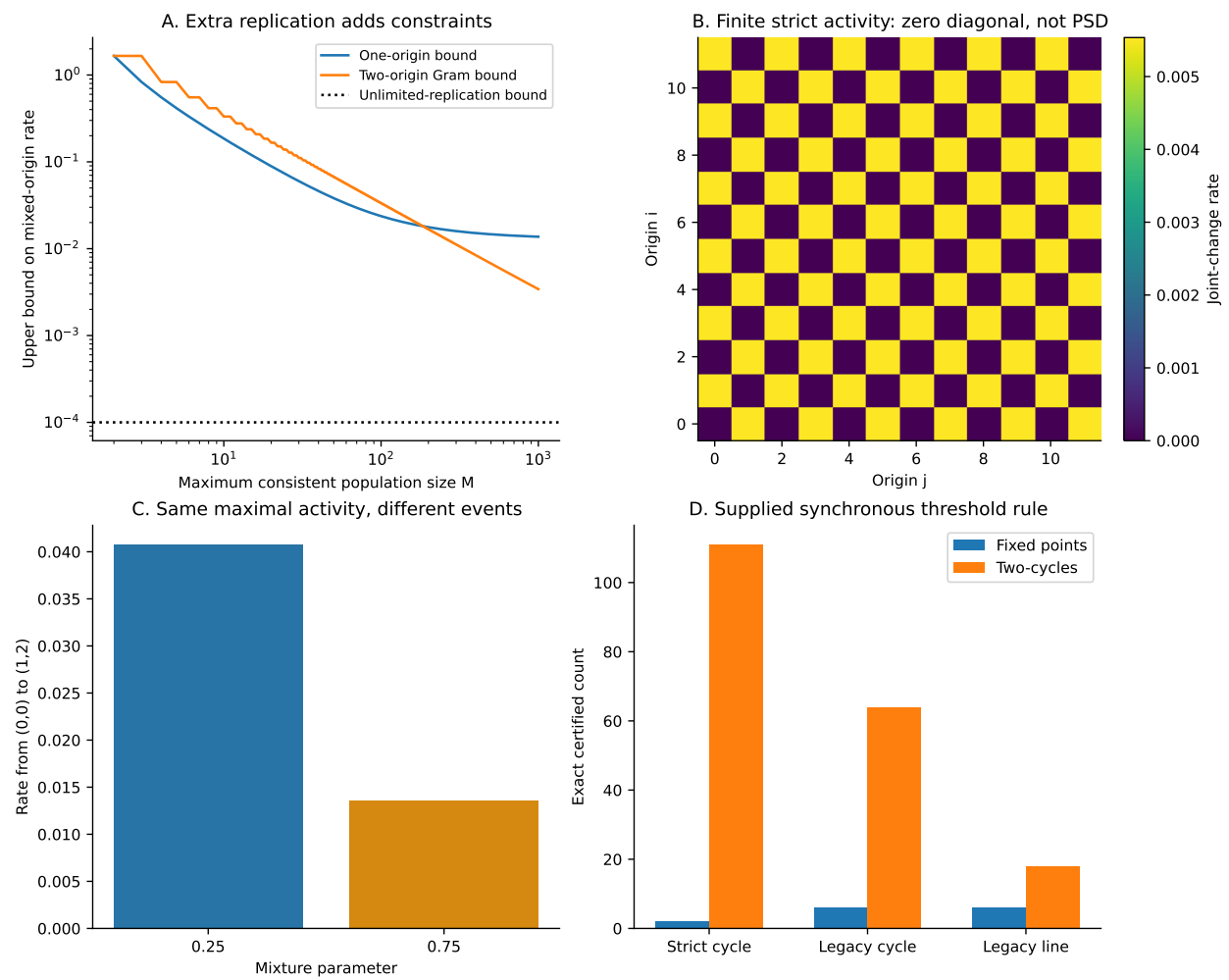
The most promising next object is not another scalar extracted from A . It is a *source-defined transition-resolved law of composition*. This campaign shows both how such a law would be constrained and how seemingly strong substitutes—finite consistency, static equilibrium, maximal activity, a neural analogy—can fail to select it. A valid nonzero-coupling prediction would be as informative as a valid independence theorem. The present evidence selects neither as fundamental physics.

No selector/QW-2191 closure, legacy completion or role transfer, physical population, clock or unit, laboratory realization, spacetime continuum, Standard Model, gravity, L_{total} or ToE closure is exported.

References

- [1] H. Crane and S. P. Lalley, *Exchangeable Markov Processes on $[k]^{\mathbb{N}}$ with Cadlag Sample Paths*, arXiv:1307.1713 (2013). Author preprint.
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A Numerical and certified comparisons



All panels describe declared mathematical models, not physical data.